Number Theory

A detailed guide: divisibility, congruences, Euler and RSA

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Companion to the *Number Theory* video series in `manim-clases` · reference: Rosen, ch. 4

How to use this guide. Every topic follows the same order: **the question** it answers, the **intuition** (what the videos show), the **formal statement**, the full **proof**, and a **worked example** line by line. Each topic ends with three memory anchors: the *proof sketch*, the *canonical counterexample* (what breaks if you drop a hypothesis) and a *rule of thumb*. Each module ends with exercises and their solutions.

Every number in this guide was checked with Python (`pow`, `math.gcd` and exhaustive search).

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1 Module 1: Divisibility, the division algorithm, gcd and the FTA

1.1 Divisibility

QUESTION. *What exactly does it mean for one number to "fit" a whole number of times into another, and which rules can you use without thinking twice?*

Definition 1.1. Let $a, b \in \mathbb{Z}$ with $a \neq 0$. We say that a **divides** b, and write $a \mid b$, if there is $k \in \mathbb{Z}$ with $b = a \cdot k$. Otherwise we write $a \nmid b$.

INTUITION. $a \mid b$ means that b can be covered exactly by blocks of size $|a|$, with nothing left over. For example, $3 \mid 12$ because 12 is four blocks of 3, and $5 \nmid 12$ because blocks of 5 leave 2 over. The whole theory in this guide comes from a single idea: *write what you know as "$b = ak$" and do algebra.*

Theorem 1.2 (Basic properties)**.** *For all $a, b, c, x, y \in \mathbb{Z}$ (with nonzero divisors):*

1. *$1 \mid a$, $a \mid 0$ and, if $a \neq 0$, $a \mid a$.*
2. ***Transitivity:** if $a \mid b$ and $b \mid c$ then $a \mid c$.*
3. ***Linear combination:** if $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$.*
4. ***Bound:** if $b \neq 0$ and $a \mid b$ then $|a| \leq |b|$.*
5. ***Antisymmetry:** if $a, b > 0$, $a \mid b$ and $b \mid a$, then $a = b$. In $\mathbb{Z}$, $a = \pm b$.*

Proof. **(1)** $a = 1 \cdot a$, $0 = a \cdot 0$ and $a = a \cdot 1$.

(2) There are $k, \ell \in \mathbb{Z}$ with $b = ak$ and $c = b\ell$. Then $c = (ak)\ell = a(k\ell)$ and $k\ell \in \mathbb{Z}$.

(3) There are k, ℓ with $b = ak$, $c = a\ell$. Then $bx + cy = akx + a\ell y = a(kx + \ell y)$, with $kx + \ell y \in \mathbb{Z}$.

(4) $b = ak$ with $k \neq 0$ (if $k = 0$ then $b = 0$). Since k is a nonzero integer, $|k| \geq 1$, so $|b| = |a||k| \geq |a|$.

(5) By (4), $a \leq b$ and $b \leq a$, so $a = b$. In $\mathbb{Z}$, applying (4) to absolute values gives $|a| = |b|$. □

Property (3) is *the* tool of the topic: almost every proof ends by saying "d divides both terms, so it divides the sum".

CANONICAL COUNTEREXAMPLE. The converse of (3) is false: $3 \mid (1 + 2)$ but $3 \nmid 1$ and $3 \nmid 2$. Knowing that d divides a sum says nothing about each term.

RULE OF THUMB. To prove $a \mid$ (something): write "something" as $a \cdot$ (integer), or as a linear combination of things a already divides.

1.2 The division algorithm

QUESTION. *If $d \nmid a$, how do you describe exactly "what's left over"? And why is it unique?*

Theorem 1.3 (Division algorithm)**.** *Given $a \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$, there are **unique** $q, r \in \mathbb{Z}$ such that*

$$a = dq + r, \qquad 0 \leq r < d.$$

Moreover $q = \lfloor a/d \rfloor$ and $r = a \bmod d = a - d\lfloor a/d \rfloor$.

INTUITION. Mark the multiples of d on the number line: $\ldots, -2d, -d, 0, d, 2d, \ldots$. The number a lands in exactly one "drawer" $[dq,\ d(q+1))$. The quotient q says which drawer, and the remainder r says how far into the drawer it went, a distance between 0 and $d - 1$. That's why, **even when a is negative, r is never negative**: it is always measured from the multiple *to the left*.

Proof. **Existence.** Let $S = \{a - dk : k \in \mathbb{Z},\ a - dk \geq 0\}$. $S \neq \varnothing$: if $a \geq 0$ take $k = 0$; if $a < 0$ take $k = a$, and then $a - da = a(1 - d) \geq 0$ because $a < 0$ and $1 - d \leq 0$. By the well-ordering principle, S has a minimum $r = a - dq \geq 0$. If $r \geq d$, then $r - d = a - d(q + 1) \geq 0$ would be in S and smaller than r, a contradiction. Hence $0 \leq r < d$.

Uniqueness. If $a = dq + r = dq' + r'$ with $0 \leq r, r' < d$, then $d(q - q') = r' - r$. So $d \mid (r' - r)$, but $-d < r' - r < d$. The only multiple of d in that interval is 0, so $r = r'$ and, since $d \neq 0$, $q = q'$. □

Example 1.4. $a = -20$, $d = 3$. The multiples of 3 around -20 are -21 and -18. The one to the left is $-21 = 3 \cdot (-7)$, and from -21 to -20 is 1. So $-20 = 3(-7) + 1$: $q = -7$, $r = 1$.



COMMON MISTAKE. $-20 = 3(-6) - 2$ is a true equation but it is **not** the division, because $r = -2 < 0$. If you get a negative remainder, subtract 1 from the quotient and add d to the remainder: $(-6, -2) \to (-7, 1)$.

1.3 Greatest common divisor

Definition 1.5. For $a, b \in \mathbb{Z}$ not both zero, $\gcd(a, b) = \max\{d \in \mathbb{Z}^+ : d \mid a \ \wedge \ d \mid b\}$. The maximum exists because the set is nonempty (1 is in it) and bounded (by the bound property, $d \leq \max(|a|, |b|)$). We say a and b are **relatively prime** (coprime) if $\gcd(a, b) = 1$.

QUESTION. *How do you compute the gcd without factoring? Factoring large numbers is hard, but dividing is easy.*

Lemma 1.6 (Euclid's lemma for the gcd)**.** *If $a = bq + r$, then* $\gcd(a, b) = \gcd(b, r)$.

Proof. The idea is to prove something stronger: *both pairs have exactly the same common divisors.* Then their maximum is the same.

- If $d \mid a$ and $d \mid b$, then $d \mid a - bq = r$ (linear combination). So d is a common divisor of b and r.
- If $d \mid b$ and $d \mid r$, then $d \mid bq + r = a$. So d is a common divisor of a and b.

The two sets of common divisors are equal, so their maximums agree. □

A useful special case is $q = 1$: $\gcd(x, y) = \gcd(x, y - x)$. It can also be proved with the double inequality $d_1 \leq d_2$ and $d_2 \leq d_1$, which is the same idea seen through the maximums.

1.4 Euclid's algorithm

INTUITION. Draw an $a \times b$ rectangle. Cut out as many squares of side b as fit (that's q_1), and what's left is a $b \times r_1$ rectangle. Repeat with that rectangle. The side of the last square, the one that fills the rectangle exactly, is the gcd: the largest square tile that tiles the original rectangle. This is exactly what the Euclid chapter of the module 1 video shows.

Theorem 1.7. *Applying successive divisions*

$$\begin{aligned} a &= bq_1 + r_1, && 0 \leq r_1 < b \\ b &= r_1 q_2 + r_2, && 0 \leq r_2 < r_1 \\ &\vdots \\ r_{k-2} &= r_{k-1} q_k + r_k, && 0 < r_k < r_{k-1} \\ r_{k-1} &= r_k q_{k+1} + 0, \end{aligned}$$

the process stops and $\gcd(a, b) = r_k$*, the* ***last nonzero remainder****.*

Proof. It stops because $b > r_1 > r_2 > \cdots \geq 0$ is a strictly decreasing sequence of natural numbers, which can't be infinite. By the lemma, $\gcd(a,b) = \gcd(b,r_1) = \cdots = \gcd(r_{k-1}, r_k) = \gcd(r_k, 0) = r_k$. □

Example 1.8. $\gcd(89, 55)$:

$$\begin{aligned} 89 &= 1 \cdot 55 + 34 \\ 55 &= 1 \cdot 34 + 21 \\ 34 &= 1 \cdot 21 + 13 \\ 21 &= 1 \cdot 13 + 8 \\ 13 &= 1 \cdot 8 + 5 \\ 8 &= 1 \cdot 5 + 3 \\ 5 &= 1 \cdot 3 + 2 \\ 3 &= 1 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 + 0 \end{aligned} \qquad \Rightarrow \qquad \gcd(89, 55) = 1.$$

Every quotient is 1 because 89 and 55 are consecutive Fibonacci numbers. That is the algorithm's *worst case*: the remainders shrink as slowly as possible (Lamé's theorem).

1.5 The Fundamental Theorem of Arithmetic

Theorem 1.9 (FTA)**.** *Every integer $n > 1$ can be written as a product of primes, $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, and that factorization is unique up to order.*

Proof idea. Existence, by strong induction: if n is prime we're done. Otherwise $n = ab$ with $1 < a, b < n$, and each factor factors by the induction hypothesis. *Uniqueness*: if $p_1 \cdots p_r = q_1 \cdots q_s$, then p_1 divides the product of the q_j. By **Euclid's lemma** (Module 2: $p \mid ab \Rightarrow p \mid a$ or $p \mid b$), p_1 divides some q_j, and since q_j is prime, $p_1 = q_j$. Cancel and repeat. □

Theorem 1.10 (Euclid: there are infinitely many primes)**.**

Proof. Suppose there are only finitely many primes $p_1, \ldots, p_k$. Let $N = p_1 p_2 \cdots p_k + 1 > 1$. By the FTA, N has some prime factor p, which must be one of the p_i. Then $p \mid N$ and $p \mid p_1 \cdots p_k$, so $p \mid N - p_1 \cdots p_k = 1$. That's impossible because $p > 1$. □

Canonical counterexample. The proof does **not** say that N is prime. $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \cdot 509$. What it says is that the prime factors of N are *new*, that is, not on the list.

gcd and lcm from the factorization. If $a = \prod p_i^{\alpha_i}$ and $b = \prod p_i^{\beta_i}$ (allowing exponents 0):

$$\gcd(a,b) = \prod p_i^{\min(\alpha_i, \beta_i)}, \qquad \operatorname{lcm}(a,b) = \prod p_i^{\max(\alpha_i, \beta_i)}, \qquad a \cdot b = \gcd(a,b) \cdot \operatorname{lcm}(a,b).$$

The last identity comes from $\min(\alpha, \beta) + \max(\alpha, \beta) = \alpha + \beta$ for each prime.

Example 1.11. $a = 2^3 \cdot 3^2 \cdot 7 = 504$, $b = 2 \cdot 3^4 \cdot 5 = 810$. Then $\gcd = 2^1 \cdot 3^2 = 18$ and $\operatorname{lcm} = 2^3 \cdot 3^4 \cdot 5 \cdot 7 = 22680$. Check: $504 \cdot 810 = 408240 = 18 \cdot 22680$.

Proof sketch. **gcd by Euclid:** (1) divide; (2) replace the pair (a, b) by (b, r) because they have the same common divisors; (3) the remainders strictly decrease, so the process stops; (4) $\gcd(r_k, 0) = r_k$.

Rule of thumb. Large numbers? Euclid. Already factored? Smallest exponents for the gcd and largest for the lcm.

Module 1 exercises

1.1 Find q and r for $a = -47$, $d = 6$.
Sol. The multiple of 6 to the left of -47 is $-48 = 6(-8)$. So $-47 = 6(-8) + 1$: $q = -8$, $r = 1$.

1.2 Compute $\gcd(252, 198)$ with Euclid.
Sol. $252 = 1 \cdot 198 + 54$, $198 = 3 \cdot 54 + 36$, $54 = 1 \cdot 36 + 18$, $36 = 2 \cdot 18 + 0$. The gcd is 18.

1.3 If $a \mid b$ and $a \mid b + 2$, prove that $a \mid 2$.
Sol. By linear combination, $a \mid (b + 2) \cdot 1 + b \cdot (-1) = 2$.

1.4 Prove that $\gcd(n, n + 1) = 1$ for every n.
Sol. If $d \mid n$ and $d \mid n + 1$, then $d \mid (n + 1) - n = 1$, so $d = 1$.

2 Module 2: Bézout, extended Euclid and modular inverses

2.1 Bézout's identity

QUESTION. *The gcd was defined as "the greatest common divisor", a* multiplicative *notion. Can it also be described* additively*, with sums of multiples of a and b?*

INTUITION. Picture two rulers with no markings, one a cm long and one b cm long. Laying them end to end or subtracting one from the other, which lengths can you measure? Exactly the numbers $sa + tb$. With $a = 6$ and $b = 10$ you can only measure multiples of 2, and you can measure 2 itself: $2 = 2 \cdot 6 - 1 \cdot 10$. The smallest length you can measure is the gcd, and every other one is a multiple of it. In the video this is a grid of points (s, t), each with its value $sa + tb$.

Theorem 2.1 (Bézout)**.** *If $a, b \in \mathbb{Z}$ are not both zero, there are $s, t \in \mathbb{Z}$ with* $\gcd(a, b) = sa + tb$*. Moreover,* $\gcd(a, b)$ *is the* ***smallest*** *positive element of the set $L = \{sa + tb : s, t \in \mathbb{Z}\}$.*

Proof. L has positive elements (for instance $a \cdot a + b \cdot b > 0$). Let $g = s_0a + t_0b$ be the smallest of them (well-ordering).

g divides a. Divide: $a = gq + r$ with $0 \leq r < g$. Then $r = a - gq = a - (s_0a + t_0b)q = (1 - s_0q)a + (-t_0q)b \in L$. If $r > 0$, it would be a positive element of L smaller than g, which is impossible. So $r = 0$ and $g \mid a$. The same argument gives $g \mid b$.

g is the greatest. If $d \mid a$ and $d \mid b$, then $d \mid s_0a + t_0b = g$, so $d \leq g$.

So g is a common divisor and is at least as large as any other: $g = \gcd(a, b)$. □

Corollary 2.2 (Coprimality test)**.** $\gcd(a, b) = 1 \iff$ *there are $s, t \in \mathbb{Z}$ with $sa + tb = 1$.*

Proof. ($\Rightarrow$) is Bézout. ($\Leftarrow$) If $d \mid a$ and $d \mid b$, then $d \mid sa + tb = 1$, so $d = 1$. □



COMMON MISTAKE. For an arbitrary g, $sa + tb = g$ does **not** imply $g = \gcd(a, b)$: $1 \cdot 6 + 1 \cdot 10 = 16$, while $\gcd(6, 10) = 2$. All you can conclude is $\gcd(a, b) \mid g$. The converse only works for $g = 1$.

2.2 The three lemmas that follow from Bézout

Lemma 2.3 (Coprime divisibility)**.** *If $a \mid bc$ and* $\gcd(a, b) = 1$*, then $a \mid c$.*

Proof (the template).

1. Since $\gcd(a, b) = 1$, by Bézout there are s, t with $sa + tb = 1$.
2. Multiply by c: $sac + tbc = c$.
3. $a \mid sac$ because $sac = a(sc)$, and $a \mid tbc$ because $bc = ak$ by hypothesis, so $tbc = a(tk)$.
4. By linear combination, $a \mid sac + tbc = c$. □

A typical exercise is solved exactly the same way, with d in place of a: $dx + by = 1 \Rightarrow adx + aby = a \Rightarrow d(ax + ky) = a$.

CANONICAL COUNTEREXAMPLE. Without coprimality it fails: $6 \mid 4 \cdot 3 = 12$ but $6 \nmid 3$. The factor 2 of 6 "hid" in the 4.

Lemma 2.4 (Euclid's lemma)**.** *If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. If $p \mid a$ there is nothing to prove. If $p \nmid a$, the only positive divisors of p are 1 and p, and p doesn't divide a, so $\gcd(p, a) = 1$. By the previous lemma, $p \mid b$. □

CANONICAL COUNTEREXAMPLE. It fails for composites: $6 \mid 2 \cdot 3$ but $6 \nmid 2$ and $6 \nmid 3$. This lemma is what makes primes special, and it is the key step in the uniqueness part of the FTA and in the self-inverse elements of Module 4.

Lemma 2.5 (Coprime divisors)**.** *If $a \mid c$, $b \mid c$ and $\gcd(a, b) = 1$, then $ab \mid c$.*

Proof. $c = ak$. Since $b \mid ak$ and $\gcd(b, a) = 1$, by Lemma 2.3 $b \mid k$, that is, $k = b\ell$. Then $c = ab\ell$. □

CANONICAL COUNTEREXAMPLE. $4 \mid 12$ and $6 \mid 12$, but $24 \nmid 12$. They share the factor 2 and you would be counting it twice.

Lemma 2.6 (gcd with a product)**.** *If $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$.*

Proof. By Bézout, $ax_1 + by_1 = 1$ and $ax_2 + cy_2 = 1$. Multiply the two identities:

$$1 = (ax_1 + by_1)(ax_2 + cy_2) = a\underbrace{(ax_1x_2 + x_1cy_2 + by_1x_2)}_{X} + (bc)\underbrace{(y_1y_2)}_{Y}.$$

Since $X, Y \in \mathbb{Z}$, the coprimality test (direction $\Leftarrow$) gives $\gcd(a, bc) = 1$. □

The idea of *multiplying two Bézout identities* works whenever you need to prove coprimality with a product.

2.3 The extended Euclidean algorithm

QUESTION. *Bézout says s, t exist. How do you find them?*

Method 1: back-substitution. Solve each division for its remainder and substitute from the bottom up, always keeping the expression as a combination of the *two most recent remainders*.

Example 2.7 (1 as a combination of 89 and 55)**.** Solving for the remainders in the table from the previous section:

$$\begin{aligned}
1 &= 3 - 1 \cdot 2 \\
&= 3 - (5 - 1 \cdot 3) = 2 \cdot 3 - 5 \\
&= 2(8 - 5) - 5 = 2 \cdot 8 - 3 \cdot 5 \\
&= 2 \cdot 8 - 3(13 - 8) = 5 \cdot 8 - 3 \cdot 13 \\
&= 5(21 - 13) - 3 \cdot 13 = 5 \cdot 21 - 8 \cdot 13 \\
&= 5 \cdot 21 - 8(34 - 21) = 13 \cdot 21 - 8 \cdot 34 \\
&= 13(55 - 34) - 8 \cdot 34 = 13 \cdot 55 - 21 \cdot 34 \\
&= 13 \cdot 55 - 21(89 - 55) = 34 \cdot 55 - 21 \cdot 89.
\end{aligned}$$

Check: $34 \cdot 55 = 1870$, $21 \cdot 89 = 1869$ and $1870 - 1869 = 1$. Another valid answer is $1 = 34 \cdot 89 - 55 \cdot 55$ $(3026 - 3025 = 1)$: Bézout coefficients are **not unique**. If (s, t) works, so does $(s + kb/g,\ t - ka/g)$. Both versions give the same inverse: 34 and $-55 \equiv 34 \pmod{89}$.

Method 2: a table with a recurrence. Start with $s_0 = 1, t_0 = 0$ (the row of a) and $s_1 = 0, t_1 = 1$ (the row of b), and at each step

$$r_j = r_{j-2} - q_{j-1}r_{j-1}, \qquad s_j = s_{j-2} - q_{j-1}s_{j-1}, \qquad t_j = t_{j-2} - q_{j-1}t_{j-1}.$$

Why it works: every row keeps the invariant $r_j = s_j a + t_j b$. The first two rows satisfy it ($a = 1a + 0b$, $b = 0a + 1b$), and if two rows satisfy it, so does the combination "$\text{row}_{j-2} - q \cdot \text{row}_{j-1}$", because everything is linear.

j	r_j	q_j	s_j	t_j	check $s_j \cdot 89 + t_j \cdot 55$
0	89	–	1	0	89
1	55	1	0	1	55
2	34	1	1	-1	34
3	21	1	-1	2	21
4	13	1	2	-3	13
5	8	1	-3	5	8
6	5	1	5	-8	5
7	3	1	-8	13	3
8	2	2	13	-21	2
9	**1**	–	$\mathbf{-21}$	**34**	1

(The q_j column is the quotient used to build the next row.)

2.4 Modular inverses

Definition 2.8. $\bar{a}$ is an **inverse of** a **modulo** m if $a\bar{a} \equiv 1 \pmod{m}$.

Theorem 2.9. *a has an inverse modulo m $\iff$ $\gcd(a, m) = 1$. In that case the inverse is unique modulo m.*

Proof. $a\bar{a} \equiv 1 \pmod{m} \iff m \mid a\bar{a} - 1 \iff a\bar{a} - 1 = mk \iff \bar{a}\,a + (-k)m = 1$ for some k. By the coprimality test, that happens if and only if $\gcd(a, m) = 1$.

Uniqueness: if $a\bar{a} \equiv a\bar{a}' \equiv 1$, then $\bar{a} = \bar{a}(a\bar{a}') = (\bar{a}a)\bar{a}' \equiv \bar{a}'$. □

Computing it: extended Euclid gives $sa + tm = 1$. Reducing mod m, the term tm disappears, so $\bar{a} \equiv s \pmod{m}$.

Example 2.10. From $34 \cdot 55 - 21 \cdot 89 = 1$, mod 89: $34 \cdot 55 \equiv 1$. So $55^{-1} \equiv 34 \pmod{89}$.

Canonical counterexample. 2 has no inverse mod 6. The row of 2 in the multiplication table mod 6 is $0, 2, 4, 0, 2, 4$, and 1 never shows up. That's because $2x - 1$ is odd and $6x'$ is even.

Rule of thumb. Does a^{-1} mod m exist? Compute $\gcd(a, m)$. If it's 1, read off the coefficient of a in Bézout and reduce it mod m.

Module 2 exercises

2.1 Write $18 = \gcd(252, 198)$ as a linear combination.
Sol. $18 = 54 - 36 = 54 - (198 - 3 \cdot 54) = 4 \cdot 54 - 198 = 4(252 - 198) - 198 = 4 \cdot 252 - 5 \cdot 198$.

2.2 Find 17^{-1} mod 60.
Sol. $60 = 3 \cdot 17 + 9$, $17 = 1 \cdot 9 + 8$, $9 = 1 \cdot 8 + 1$. Back-substituting: $1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(60 - 3 \cdot 17) - 17 = 2 \cdot 60 - 7 \cdot 17$. So $17^{-1} \equiv -7 \equiv 53$. Check: $17 \cdot 53 = 901 = 15 \cdot 60 + 1$.

2.3 Prove that if $\gcd(a, b) = 1$ then $\gcd(a + b, b) = 1$.
Sol. $sa + tb = 1 \Rightarrow s(a + b) + (t - s)b = 1$.

2.4 Does 15 have an inverse mod 42? *Sol.* No, because $\gcd(15, 42) = 3 \neq 1$.

3 Module 3: Congruences, the CRT, Fermat and Wilson

3.1 The congruence relation

Definition 3.1. $a \equiv b \pmod m \iff m \mid (a-b) \iff a \bmod m = b \bmod m$.

Intuition. It's clock arithmetic: on a clock with m hours, a and b show the same hour. Visually, coil the number line into a spring with m positions per turn. Numbers that end up stacked on top of each other are congruent, and each "column" of the spring is a **class** $[r]_m = \{r + km : k \in \mathbb{Z}\}$.

Theorem 3.2. $\equiv \pmod m$ *is an equivalence relation and splits* $\mathbb{Z}$ *into exactly* m *classes* $[0], [1], \ldots, [m-1]$.

Proof. Reflexive: $m \mid 0$. *Symmetric:* $m \mid a-b \Rightarrow m \mid b-a$. *Transitive:* $m \mid a-b$ and $m \mid b-c \Rightarrow m \mid (a-b)+(b-c) = a-c$. There are m classes because, by the division algorithm, each integer has a unique remainder in $\{0, \ldots, m-1\}$. □

3.2 Algebra of congruences

Theorem 3.3. *If* $a \equiv b$ *and* $c \equiv d \pmod m$*, then* $a+c \equiv b+d$*,* $ac \equiv bd$*,* $a^k \equiv b^k$ *and* $P(a) \equiv P(b)$ *for every polynomial* P *with integer coefficients.*

Proof. Sums: $(a+c)-(b+d) = (a-b)+(c-d)$. Products, with the *add and subtract* trick: $ac - bd = ac - bc + bc - bd = c(a-b) + b(c-d)$, and m divides both terms. Powers follow by induction from products, and polynomials by combining sums and products. □

In practice: you can reduce mod m *at any point* in a computation with sums, products and powers. For example, $27 \equiv 9 \cdot 3 \equiv 4 \cdot 3 = 12 \equiv 2 \pmod 5$.

Question. *Adding, multiplying and raising to powers always work. What about* dividing*?*

Theorem 3.4 (Cancellation)**.** $ac \equiv bc \pmod m \implies a \equiv b \pmod{\frac{m}{\gcd(c,m)}}$*. In particular, if* $\gcd(c,m) = 1$ *you can cancel without changing the modulus.*

Proof. Let $g = \gcd(c,m)$, with $c = gc'$, $m = gm'$ and $\gcd(c',m') = 1$. We have $m \mid c(a-b)$, that is, $gm' \mid gc'(a-b)$, so $m' \mid c'(a-b)$. Since $\gcd(m',c') = 1$, by the coprime divisibility lemma $m' \mid a-b$. □



Common mistake. $4x \equiv 4y \pmod 6$ does **not** imply $x \equiv y \pmod 6$. Counterexample: $x = 0$, $y = 3$. Then $0 \equiv 12 \pmod 6$ but $0 \not\equiv 3 \pmod 6$. The correct conclusion is $x \equiv y \pmod 3$, because $6/\gcd(4,6) = 3$.

3.3 Linear congruences $ax \equiv b \pmod m$

Theorem 3.5. *Let* $d = \gcd(a,m)$*. The congruence* $ax \equiv b \pmod m$ *has a solution* $\iff d \mid b$*. In that case it has* ***exactly*** d *solutions that are incongruent mod* m*:*

$$x \equiv x_0 + k \cdot \tfrac{m}{d} \pmod m, \qquad k = 0, 1, \ldots, d-1,$$

where x_0 *solves* $\frac{a}{d}x \equiv \frac{b}{d} \pmod{\frac{m}{d}}$.

Proof. Necessity: if $ax - b = mk$, then $b = ax - mk$, and d divides a and m, so $d \mid b$.

Sufficiency: if $d \mid b$, divide everything by d: $a'x \equiv b' \pmod{m'}$ with $a' = a/d$, $b' = b/d$, $m' = m/d$ and $\gcd(a', m') = 1$. Then a' has an inverse and $x_0 \equiv (a')^{-1}b' \pmod{m'}$. The solutions mod m' are $x_0 + km'$. Mod $m = dm'$, the values $k = 0, \ldots, d-1$ give distinct remainders (they're less than m apart), and $k = d$ comes back to x_0. □

INTUITION. If $d > 1$, the "orbit" $\{ax \bmod m\}$ only visits multiples of d, and visits each of them d times. If b isn't a multiple of d, you never reach it. If it is, you reach it d times.

Example 3.6 ($55x \equiv 13 \pmod{89}$)**.** $\gcd(55, 89) = 1$, so there is a single solution. Multiply by $55^{-1} \equiv 34$: $x \equiv 13 \cdot 34 = 442 = 4 \cdot 89 + 86$, so $x \equiv 86 \pmod{89}$. Check: $55 \cdot 86 = 4730 = 53 \cdot 89 + 13$.

Example 3.7 (With $d > 1$: $6x \equiv 9 \pmod{15}$)**.** $d = \gcd(6, 15) = 3$ and $3 \mid 9$, so there are 3 solutions. Dividing by 3: $2x \equiv 3 \pmod 5$. Since $2^{-1} \equiv 3 \pmod 5$, $x_0 \equiv 9 \equiv 4 \pmod 5$. The solutions mod 15 are 4, 9, 14. Check: $24, 54, 84$ all leave remainder 9 when divided by 15.

3.4 The Chinese Remainder Theorem

QUESTION. *If I know the remainder of x mod 3 and mod 5, do I know x mod 15?*

INTUITION. Place the numbers $0, \ldots, 14$ on a 3×5 grid, each in row x mod 3 and column x mod 5. Going up by one moves you down a row and over a column, diagonally, wrapping around the edges. Since 3 and 5 are coprime, the diagonal *visits all 15 cells without repeating.* So each pair (remainder mod 3, remainder mod 5) corresponds to exactly one number mod 15. With 2×4 (not coprime) the diagonal closes after only $\operatorname{lcm}(2, 4) = 4$ steps and leaves cells empty: some systems have no solution. This grid is the central animation of module 3.

Theorem 3.8 (CRT)**.** *If $m_1, \ldots, m_n$ are pairwise coprime, the system $x \equiv a_i \pmod{m_i}$ ($i = 1, \ldots, n$) has a unique solution modulo $M = m_1 \cdots m_n$.*

Proof. Existence (construction). Let $M_i = M/m_i$. Since M_i is a product of moduli coprime to m_i, we have $\gcd(M_i, m_i) = 1$ (by the gcd-with-a-product lemma, applied repeatedly). So $y_i = M_i^{-1} \bmod m_i$ exists. Let

$$x = \sum_{i=1}^{n} a_i M_i y_i.$$

Mod m_j: if $i \neq j$, $m_j \mid M_i$ and that term is $\equiv 0$. The j-th term gives $a_j M_j y_j \equiv a_j \cdot 1$. So $x \equiv a_j \pmod{m_j}$ for every j.

Uniqueness. If x and x' are solutions, $m_i \mid x - x'$ for every i. By the coprime divisors lemma, applied repeatedly, $M \mid x - x'$. □

INTUITION. Each term $C_i = M_i y_i$ is a "switch": it is 1 mod m_i and 0 mod all the others. The solution is the sum of the remainders, each one multiplied by its switch.

Example 3.9 (By substitution)**.** $x \equiv 5 \pmod{55}$, $x \equiv 10 \pmod{89}$. From the first equation, $x = 55k + 5$. In the second, $55k \equiv 5 \pmod{89}$, so $k \equiv 5 \cdot 34 = 170 \equiv 81 \pmod{89}$. With $k = 89q + 81$: $x = 55(89q + 81) + 5 = 4895q + 4460$. So $x \equiv 4460 \pmod{4895}$.

Example 3.10 (With the formula)**.** $2x \equiv 1 \pmod 5$, $x \equiv -1 \pmod 7$, $x \equiv 4 \pmod 9$, $3x \equiv 2 \pmod{11}$. **First solve each one for x:** $2^{-1} \equiv 3 \pmod 5$, so $x \equiv 3$. $-1 \equiv 6 \pmod 7$. $3^{-1} \equiv 4 \pmod{11}$, so $x \equiv 8$. The system becomes $x \equiv 3, 6, 4, 8 \bmod 5, 7, 9, 11$, with $M = 3465$:

m_i	a_i	M_i	$M_i \bmod m_i$	$y_i = M_i^{-1}$
5	3	693	3	2
7	6	495	5	3
9	4	385	7	4
11	8	315	7	8

$x \equiv 3{\cdot}693{\cdot}2{+}6{\cdot}495{\cdot}3{+}4{\cdot}385{\cdot}4{+}8{\cdot}315{\cdot}8 = 4158{+}8910{+}6160{+}20160 = 39388 \equiv 1273 \pmod{3465}$. Check: $1273 = 5 \cdot 254 + 3 = 7 \cdot 181 + 6 = 9 \cdot 141 + 4 = 11 \cdot 115 + 8$.

Example 3.11 (Moduli that are *not* coprime)**.** When the moduli share factors, **split each equation into prime powers** (the CRT used backwards):

- $\{x \equiv 7 \pmod{10},\ x \equiv 4 \pmod{35}\}$ gives $x \equiv 1 \pmod 2$, $x \equiv 2 \pmod 5$, $x \equiv 4 \pmod 5$, $x \equiv 4 \pmod 7$. There are two contradictory conditions mod 5, so **there is no solution**.
- $\{x \equiv 7 \pmod{10},\ x \equiv 2 \pmod{35}\}$ gives $x \equiv 1 \pmod 2$, $x \equiv 2 \pmod 5$ (which appears twice, without conflict) and $x \equiv 2 \pmod 7$. With the CRT and $M = 70$: $C_1 = 35{\cdot}1$, $C_2 = 14{\cdot}4 = 56$, $C_3 = 10 \cdot 5 = 50$. So $x \equiv 35 + 2 \cdot 56 + 2 \cdot 50 = 247 \equiv 37 \pmod{70}$.

General rule: $x \equiv a \pmod m$, $x \equiv b \pmod n$ has a solution $\iff \gcd(m, n) \mid a - b$, and it is unique mod $\operatorname{lcm}(m, n)$, not mod mn.

Proof sketch. **CRT:** (1) $\gcd(M_i, m_i) = 1$, so y_i exists; (2) $C_i \equiv 1 \pmod{m_i}$ and $C_i \equiv 0$ for the other moduli; (3) add up $\sum a_i C_i$; (4) uniqueness: if every m_i divides $x - x'$, so does their product.

3.5 Fermat's Little Theorem

Question. *How do you compute* $2^{1000005} \bmod 101$ *without a million multiplications?*

Theorem 3.12 (FLT)**.** *If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod p$. For every $a \in \mathbb{Z}$, $a^p \equiv a \pmod p$.*

Intuition. Multiplying by a *shuffles* the remainders $1, 2, \ldots, p-1$ without repeating or losing any. With $p = 7$ and $a = 3$: $3, 6, 2, 5, 1, 4$ are the same six numbers in a different order. If you multiply them all, the product is the same before and after the shuffle, but after the shuffle there is an extra factor a^{p-1}.

Proof. Consider $a, 2a, \ldots, (p-1)a \bmod p$.

- None is $\equiv 0$: if $p \mid ka$ with $p \nmid a$, by Euclid's lemma $p \mid k$, which is impossible since $1 \le k \le p - 1$.
- They are distinct: if $ia \equiv ja$, cancel a (because $\gcd(a, p) = 1$) and get $i \equiv j$.

They are $p - 1$ distinct nonzero remainders, so they are $1, \ldots, p - 1$ in some order. Multiplying everything:

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod p.$$

Since $\gcd((p-1)!, p) = 1$, cancel it and $a^{p-1} \equiv 1$ is left. For the second form: if $p \nmid a$ multiply by a, and if $p \mid a$ both sides are $\equiv 0$. □

Strategy for huge exponents. Divide the exponent by $p - 1$: $E = q(p-1) + r$, and then $a^E = (a^{p-1})^q a^r \equiv a^r \pmod p$. *The exponent lives mod $p - 1$.*

Example 3.13. $2^{1000005} \bmod 101$. Since 101 is prime and $101 \nmid 2$, $2^{100} \equiv 1$. Also $1000005 = 100 \cdot 10000 + 5$, so $2^{1000005} \equiv 2^5 = 32 \pmod{101}$.

Example 3.14 (An inverse via Fermat)**.** $7^{22} \equiv 1 \pmod{23}$, so $7 \cdot 7^{21} \equiv 1$ and $7^{-1} \equiv 7^{21}$. By repeated squaring: $7^2 = 49 \equiv 3$, $7^4 \equiv 9$, $7^8 \equiv 81 \equiv 12$, $7^{16} \equiv 144 \equiv 6$. Since $21 = 16 + 4 + 1$: $7^{21} \equiv 6 \cdot 9 \cdot 7 = 378 = 16 \cdot 23 + 10 \equiv 10$. Check: $7 \cdot 10 = 70 = 3 \cdot 23 + 1$.



COMMON MISTAKE. Fermat **only** holds for a prime modulus. For $a^{12} \bmod 15$ the exponent 14 is useless. Use Euler ($\varphi(15) = 8$) or the CRT with moduli 3 and 5.

CANONICAL COUNTEREXAMPLE. The converse is false: $2^{340} \equiv 1 \pmod{341}$ and $341 = 11 \cdot 31$ is not prime (it's a pseudoprime). $a^{n-1} \equiv 1$ does not prove that n is prime.

3.6 Wilson's Theorem

Theorem 3.15 (Wilson)**.** $n > 1$ *is prime* $\iff (n-1)! \equiv -1 \pmod{n}$.

INTUITION. In $\{1, \ldots, p-1\}$ every number has an inverse, and almost all of them pair up with a *different* number: the pair $a \cdot a^{-1} = 1$ cancels from the product. The only numbers that are their own inverse are the solutions of $a^2 \equiv 1$, that is, 1 and $p-1$ (see Module 4). So the whole product is $1 \cdot (p-1) \equiv -1$. With $p = 7$: $6! = 1 \cdot (2 \cdot 4)(3 \cdot 5) \cdot 6 \equiv 1 \cdot 1 \cdot 1 \cdot 6 \equiv -1$.

Proof. ($\Rightarrow$) For $p = 2$: $1! \equiv -1 \pmod{2}$. For odd p: every $a \in \{2, \ldots, p-2\}$ has an inverse $a^{-1} \neq a$, because $a^2 \equiv 1 \iff p \mid (a-1)(a+1) \iff a \equiv \pm 1$ by Euclid's lemma. So $\{2, \ldots, p-2\}$ splits into pairs whose product is 1, and then $(p-1)! \equiv 1 \cdot 1 \cdots 1 \cdot (p-1) \equiv -1$.

($\Leftarrow$) If $n = ab$ with $1 < a < n$, then $a \mid (n-1)!$. If $(n-1)! \equiv -1 \pmod{n}$, since $a \mid n$ also $(n-1)! \equiv -1 \pmod{a}$, and then $a \mid 1$. Contradiction. □

Example 3.16. With $p = 101$: $100! \equiv -1 \equiv 100$. Since $100! = 100 \cdot 99!$ and $100 \equiv -1$: $-99! \equiv -1$, so $99! \equiv 1$. Since $99! = 99 \cdot 98!$ and $99 \equiv -2$: $-2 \cdot 98! \equiv 1$, that is, $2 \cdot 98! \equiv -1 \equiv 100$, and so $98! \equiv 50 \pmod{101}$. Check: $2 \cdot 50 = 100 \equiv -1$.

RULE OF THUMB. A factorial mod a prime p? Wilson for $(p-1)!$, then peel off factors by writing them as $-1, -2, \ldots$

Module 3 exercises

3.1 Solve $8x \equiv 12 \pmod{20}$.
Sol. $d = 4 \mid 12$, so there are 4 solutions. Dividing: $2x \equiv 3 \pmod{5}$, so $x \equiv 4 \pmod{5}$. The solutions are $4, 9, 14, 19 \pmod{20}$.

3.2 Solve $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.
Sol. $M = 105$. $C_1 = 35 \cdot (35^{-1} \bmod 3) = 35 \cdot 2 = 70$, $C_2 = 21 \cdot (21^{-1} \bmod 5) = 21 \cdot 1 = 21$, $C_3 = 15 \cdot (15^{-1} \bmod 7) = 15 \cdot 1 = 15$. So $x \equiv 140 + 63 + 30 = 233 \equiv 23 \pmod{105}$.

3.3 Compute $3^{100} \bmod 7$.
Sol. $3^6 \equiv 1$ and $100 = 6 \cdot 16 + 4$, so $3^{100} \equiv 3^4 = 81 \equiv 4$.

3.4 Compute $15! \bmod 17$.
Sol. $16! \equiv -1$ and $16 \equiv -1$, so $-15! \equiv -1$ and $15! \equiv 1$.

3.5 Does $\{x \equiv 3 \pmod{6},\ x \equiv 5 \pmod{8}\}$ have a solution?
Sol. $\gcd(6, 8) = 2 \mid 5 - 3 = 2$, so yes. The solutions are the odd numbers that are $\equiv 3 \pmod{6}$ and $\equiv 5 \pmod{8}$: $x \equiv 21 \pmod{24}$.

4 Module 4: Euler's φ function, Euler's Theorem and RSA

4.1 The φ function

Definition 4.1. $\varphi(n) = \big|\{x : 1 \le x \le n,\ \gcd(x,n) = 1\}\big|$, that is, how many remainders have an inverse mod n.

Theorem 4.2 (Formulas)**.** *1.* $\varphi(p) = p - 1$ *if* p *is prime.*

2. $\varphi(p^k) = p^k - p^{k-1} = p^k\left(1 - \frac{1}{p}\right)$.

3. If $\gcd(m,n) = 1$*:* $\varphi(mn) = \varphi(m)\varphi(n)$.

4. If $n = p_1^{a_1} \cdots p_k^{a_k}$*:* $\varphi(n) = n\prod_i \left(1 - \frac{1}{p_i}\right)$.

Proof. (1) Every number from 1 to $p-1$ is coprime to p. (2) $\gcd(x, p^k) \neq 1 \iff p \mid x$. Between 1 and p^k there are p^{k-1} multiples of p, and those are subtracted. (3) By the CRT, $x \bmod mn \leftrightarrow (x \bmod m,\ x \bmod n)$ is a bijection. Also $\gcd(x, mn) = 1 \iff \gcd(x,m) = 1$ and $\gcd(x,n) = 1$. So the invertible remainders mod mn correspond to the pairs (invertible mod m, invertible mod n), and there are $\varphi(m)\varphi(n)$ of those. On the CRT grid they're the cells that are in a "good" row and a "good" column at the same time. (4) Combine (2) and (3). □

Example 4.3. $\varphi(119) = \varphi(7)\varphi(17) = 6 \cdot 16 = 96$. $\varphi(1000) = \varphi(2^3)\varphi(5^3) = 4 \cdot 100 = 400$. $\varphi(360) = 360 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 96$.



Common mistake. $\varphi(mn) = \varphi(m)\varphi(n)$ **only** when they are coprime: $\varphi(4) = 2 \neq \varphi(2)^2 = 1$.

4.2 Euler's Theorem

Theorem 4.4 (Euler)**.** *If* $\gcd(a,n) = 1$*, then* $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Proof. It's Fermat's proof, but using only the invertible remainders. Let $r_1, \ldots, r_{\varphi(n)}$ be the remainders coprime to n. The products $ar_1, \ldots, ar_{\varphi(n)}$ are still coprime to n (by the gcd-with-a-product lemma) and are distinct (you can cancel a). So they are the same r_i in a different order. Multiplying everything: $a^{\varphi(n)} \prod r_i \equiv \prod r_i$, and $\prod r_i$, which is invertible, cancels. Fermat is the case $n = p$. □

Strategy: reduce the exponent mod $\varphi(n)$.

Example 4.5 ($2^{118} \bmod 119$)**.** $\varphi(119) = 96$ and $\gcd(2, 119) = 1$, so $2^{118} = 2^{96} \cdot 2^{22} \equiv 2^{22}$. Next, $2^7 = 128 \equiv 9$, $2^{11} = 2^7 \cdot 2^4 \equiv 9 \cdot 16 = 144 \equiv 25$ and $2^{22} = (2^{11})^2 \equiv 625 = 5 \cdot 119 + 30 \equiv 30$. The remainder is **30**.

Example 4.6 (The last three digits of 7^{2407})**.** The last three digits are the remainder mod 1000. $\varphi(1000) = 400$ and $2407 = 6 \cdot 400 + 7$, so $7^{2407} \equiv 7^7$. Also $7^4 = 2401 \equiv 401$ and $7^3 = 343$, so $7^7 \equiv 401 \cdot 343 = 137543 \equiv 543$. The last three digits are **543**.

4.3 Self-inverse elements: $a^2 \equiv 1$

Question. *How many remainders are their own inverse? In the reals there are only two,* ± 1*. Is it the same mod* n*?*

Theorem 4.7. *1. If* p *is an odd prime,* $a^2 \equiv 1 \pmod{p^k}$ *has exactly 2 solutions,* ± 1*. (We proved it for* p *above. For* p^k *the argument is the same, because* p *can't divide both* $a - 1$ *and* $a + 1$*, since their difference is 2.)*

2. *$a^2 \equiv 1 \pmod{2^n}$ has 1 solution if $n = 1$, 2 if $n = 2$, and 4 if $n \geq 3$: ± 1 and $2^{n-1} \pm 1$.*

3. *For general n, by the CRT, the number of solutions is the product of the numbers of solutions for each prime power.*

Proof. (1) $p \mid (a-1)(a+1)$. By Euclid's lemma, $p \mid a-1$ or $p \mid a+1$. So $a \equiv \pm 1$, and these are distinct because $p \neq 2$.

(2) With $n \geq 3$: a must be odd, so $a-1$ and $a+1$ are consecutive even numbers. By $\gcd(x, y) = \gcd(x, y-x)$: $\gcd(a-1, a+1) = \gcd(a-1, 2) = 2$. So only one of the two factors can have more than one factor of 2, and since the product needs at least n factors of 2, one of them must be divisible by 2^{n-1}:

- $2^{n-1} \mid a-1$: $a \equiv 1$ or $a \equiv 2^{n-1}+1 \pmod{2^n}$.
- $2^{n-1} \mid a+1$: $a \equiv 2^{n-1}-1$ or $a \equiv -1 \pmod{2^n}$.

For $n \geq 3$ the four values are distinct, and all four work: $(2^{n-1} \pm 1)^2 = 2^{2n-2} \pm 2^n + 1 \equiv 1$, because $2n - 2 \geq n$. □

Example 4.8 (Mod 100)**.** $100 = 4 \cdot 25$. Mod 4: $a \equiv 1, 3$. Mod 25: $a \equiv \pm 1$. There are $2 \cdot 2 = 4$ solutions: $\{1, 49, 51, 99\}$. **The product of the 40 remainders coprime to 100:** each one that isn't its own inverse cancels with its partner, leaving $1 \cdot 49 \cdot 51 \cdot 99 \equiv 1 \cdot (-51) \cdot 51 \cdot (-1) = 51^2 = 2601 \equiv 1 \pmod{100}$. This generalizes Wilson: the product of the invertible elements is -1 only when there are exactly two self-inverse elements.

Example 4.9 (Mod 3600)**.** $3600 = 2^4 \cdot 3^2 \cdot 5^2$. Mod 16: $\{1, 7, 9, 15\}$. Mod 9: $\{1, 8\}$. Mod 25: $\{1, 24\}$. Total: $4 \cdot 2 \cdot 2 = 16$. With $C_1 = 225 \cdot 1$, $C_2 = 400 \cdot 7 = 2800$ and $C_3 = 144 \cdot 4 = 576$, $a \equiv 225r_1 + 2800r_2 + 576r_3 \pmod{3600}$ gives:

$$\{1, 199, 449, 649, 1151, 1351, 1601, 1799, 1801, 1999, 2249, 2449, 2951, 3151, 3401, 3599\}.$$

(Checked by exhaustive search: these are exactly the $a < 3600$ with $a^2 \equiv 1$.)

4.4 RSA

Question. *How can anyone encrypt a message to me with a public key, so that nobody but me can decrypt it?*

Key generation.

1. Pick primes p, q. Compute $n = pq$ and $\varphi(n) = (p-1)(q-1)$.
2. Pick e with $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$.
3. Compute $d \equiv e^{-1} \pmod{\varphi(n)}$ with extended Euclid.
4. The public key is (n, e). The private key is d, and p, q and $\varphi(n)$ are kept secret.

Encrypt: $C = M^e \bmod n$. **Decrypt:** $M = C^d \bmod n$.

Theorem 4.10 (Why it works)**.** *If $0 \leq M < n$, then $(M^e)^d \equiv M \pmod{n}$.*

Proof. $ed = 1 + k\varphi(n) = 1 + k(p-1)(q-1)$. We show $M^{ed} \equiv M \pmod{p}$. If $p \mid M$, both sides are 0. If not, by Fermat $M^{ed} = M \cdot (M^{p-1})^{k(q-1)} \equiv M$. The same holds mod q. Since $\gcd(p, q) = 1$, by the CRT (uniqueness) $M^{ed} \equiv M \pmod{pq}$. □

Example 4.11 (Full RSA with small numbers)**.** $p = 5$, $q = 11$, so $n = 55$ and $\varphi(n) = 40$. Take $e = 3$, with $\gcd(3, 40) = 1$. Since $40 = 13 \cdot 3 + 1$, $1 = 40 - 13 \cdot 3$ and $d \equiv -13 \equiv 27 \pmod{40}$. Check: $3 \cdot 27 = 81 = 2 \cdot 40 + 1$.

Encrypt $M = 7$: $C = 7^3 = 343 = 6 \cdot 55 + 13$, so $C = 13$.

Decrypt $13^{27} \bmod 55$ with the CRT. Mod 5: $13 \equiv 3$, and $3^{27} = 3^{4 \cdot 6 + 3} \equiv 3^3 = 27 \equiv 2$. Mod 11: $13 \equiv 2$, and $2^{27} = 2^{10 \cdot 2 + 7} \equiv 2^7 = 128 \equiv 7$. We need $x \equiv 2 \pmod 5$, $x \equiv 7 \pmod{11}$, and the solution is $x = 7$. We get $M = 7$ back.



COMMON MISTAKE. d is the inverse of e **mod** $\varphi(n)$**, not mod** n. The reason is that exponents "live" mod $\varphi(n)$ (Euler), while bases live mod n.

RULE OF THUMB. To pick a theorem: is the modulus prime? Fermat (exponent mod $p - 1$). Not prime but $\gcd(a, n) = 1$? Euler (exponent mod $\varphi(n)$). Can the modulus be factored? The CRT, solving piece by piece. A factorial mod a prime? Wilson.

Module 4 exercises

4.1 Compute the last two digits of 3^{2026}.
Sol. $\varphi(100) = 40$ and $2026 \equiv 26 \pmod{40}$. $3^5 = 243 \equiv 43$, $3^{10} \equiv 43^2 = 1849 \equiv 49$, $3^{20} \equiv 49^2 = 2401 \equiv 1$. So $3^{26} = 3^{20} \cdot 3^5 \cdot 3 \equiv 1 \cdot 43 \cdot 3 = 129 \equiv 29$. The last two digits are **29**.

4.2 Find every a with $a^2 \equiv 1 \pmod{15}$.
Sol. Mod 3: ± 1. Mod 5: ± 1. The four combinations give $\{1, 4, 11, 14\}$.

4.3 In RSA with $p = 5$, $q = 11$ and $e = 3$, someone computes d as $3^{-1} \bmod 55 = 37$. Does it work?
Sol. Not in general: $13^{37} \bmod 55 = 18 \neq 7$. The reason is that $3 \cdot 37 = 111 \equiv 31 \pmod{40}$, which isn't 1.

4.4 Compute $\varphi(2^5 \cdot 7^2)$.
Sol. $\varphi(32)\varphi(49) = 16 \cdot 42 = 672$.

5 Decision map and common mistakes

If the problem asks for...	**use...**
$\gcd(a, b)$ of large numbers	Euclid (last nonzero remainder)
s, t with $sa + tb = \gcd$	Extended Euclid (back-substitution or the table)
$a^{-1} \bmod m$	check gcd = 1, then Bézout. If m is prime, also a^{m-2}
$ax \equiv b \pmod m$	$d = \gcd(a, m)$; if $d \mid b$, divide by d, and there are d solutions
Several congruences, coprime moduli	CRT (formula or substitution)
Several congruences, non-coprime moduli	split into prime powers and look for clashes
$a^E \bmod p$, p prime	Fermat: $E \bmod (p-1)$
$a^E \bmod n$, $\gcd(a, n) = 1$	Euler: $E \bmod \varphi(n)$
$a^E \bmod n$, n composite	also the CRT with the factors of n
$k! \bmod p$	Wilson, peeling off factors ($p - 1 \equiv -1, \dots$)
Last k digits	mod 10^k with Euler, $\varphi(10^k) = 4 \cdot 10^{k-1}$
Prove $d \mid c$ using coprimality	the Bézout template: $sa + tb = 1$, multiply by c

Common mistakes, all together

1. A negative remainder in the division algorithm: r always satisfies $0 \le r < d$.
2. $sa + tb = g$ does not imply $g = \gcd$, except when $g = 1$.
3. Cancelling without checking the gcd: $ac \equiv bc \pmod m$ gives $a \equiv b \pmod{m/\gcd(c, m)}$.
4. Fermat with a composite modulus.
5. The CRT with non-coprime moduli: there may be no solution, and if there is one it is unique mod the lcm.
6. $\varphi(mn) = \varphi(m)\varphi(n)$ without coprimality.
7. In RSA, $d = e^{-1} \bmod \varphi(n)$, not mod n.
8. In $ax \equiv b$, forgetting that there are d solutions mod m, not just one.
9. Thinking that $p_1 \cdots p_k + 1$ is prime.